

2-6 Use Partial Sums to Add

Learning goal: I can break apart addends by place value, add the partial sums, and show my work in a row or stacked.

Textbook: pages 55–58. The book adds $367 + 145$ with partial sums written in a row and stacked.

Words to Know

decompose	Break apart a number by place value: $367 = 300 + 60 + 7$.
addend	A number you add.
partial sum	The sum of one place: the hundreds, the tens, or the ones. $300 + 100 = 400$ is a partial sum.
sum	The answer when you add all the partial sums.
stack	Write the numbers one under the other, lining up the places.

How to Teach It

1. Say each addend in expanded form: 368 is $300 + 60 + 8$; 154 is $100 + 50 + 4$.
2. Add the hundreds, then the tens, then the ones. Each answer is a **partial sum**: 400, 110, 12.
3. Add the partial sums to find the sum: $400 + 110 + 12 = 522$.
4. Show it **in a row** (equations) or **stacked** (labels on the left, partial sums on the right). The sum is the same either way because the parts are the same.
5. To work backward, the first number in each partial-sum equation builds one addend, and the second number builds the other.

Tip: Ask, “What is this digit worth?” and “Did you add every partial sum?” Watch for children who stop at the hundreds or write 12 as just 2.

Worked Examples

Example 1: Partial sums in a row

$$\begin{aligned}
 368 + 154 &= 522 \\
 300 + 100 &= 400 \\
 60 + 50 &= 110 \\
 8 + 4 &= 12 \\
 400 + 110 + 12 &= \mathbf{522}
 \end{aligned}$$

Example 2: Stacked partial sums

$$\begin{array}{r}
 368 \\
 + 154 \\
 \hline
 300 + 100 \quad 400 \\
 60 + 50 \quad 110 \\
 8 + 4 \quad + 12 \\
 \hline
 \mathbf{522}
 \end{array}$$

Example 3: Work backward

$400 + 200 = 600$, $30 + 50 = 80$, $1 + 7 = 8$
 First numbers: 400, 30, 1 → **431**. Second numbers:
 200, 50, 7 → **257**.
 So the equation was $431 + 257 = 688$.

Example 4: Watch for this mistake

$406 + 237 \rightarrow$ partial sums 600, 30, 13.
 The sum is **not** 600. 600 is only the hundreds. $600 + 30 + 13 = \mathbf{643}$.

Skills Checklist *Check off each skill your child can do without help.*

- | | |
|---|---|
| ☐ Break apart a 3-digit number by place value. | ☐ Explain why both ways give the same sum. |
| ☐ Find partial sums in a row. | ☐ Find the addends from the partial sums. |
| ☐ Find partial sums stacked. | ☐ Spot a mistake, like stopping at the hundreds. |

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Name _____ Date _____

Together *Work with your grown-up. Talk about each step.*

- A.** Break apart each addend. Find the partial sums in a row. Then find the sum.

$$256 + 318 = ?$$

$$\underline{\hspace{2cm}} + \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$$

$$\underline{\hspace{2cm}} + \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$$

$$\underline{\hspace{2cm}} + \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$$

$$\underline{\hspace{2cm}} + \underline{\hspace{2cm}} + \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$$

- B.** Stack the addends. Find the partial sums. Then find the sum.

$$\begin{array}{r} 427 \\ + 165 \\ \hline \end{array}$$

$$\begin{array}{r} \\ + \\ \hline \end{array}$$

- C.** Kai adds $406 + 237$. His partial sums are 600, 30, and 13. Kai says the sum is 600. Do you agree? Explain.

On My Own *Try these by yourself. Show your work.*

- 1.** Use partial sums in a row.

$$358 + 426 = ?$$

$$\underline{\hspace{2cm}} + \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$$

$$\underline{\hspace{2cm}} + \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$$

$$\underline{\hspace{2cm}} + \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$$

$$\underline{\hspace{2cm}} + \underline{\hspace{2cm}} + \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$$

- 2.** Use stacked partial sums.

$$\begin{array}{r} 615 \\ + 273 \\ \hline \end{array}$$

$$\begin{array}{r} \\ + \\ \hline \end{array}$$

3. Use stacked partial sums.

$$\begin{array}{r} 539 \\ + 247 \\ \hline \\ \\ \\ + \\ \hline \\ \hline \end{array}$$

4. Use partial sums in a row.

$$264 + 189 = ?$$
$$\begin{array}{l} \underline{\quad\quad} + \underline{\quad\quad} = \underline{\quad\quad} \\ \underline{\quad\quad} + \underline{\quad\quad} = \underline{\quad\quad} \\ \underline{\quad\quad} + \underline{\quad\quad} = \underline{\quad\quad} \\ \underline{\quad\quad} + \underline{\quad\quad} + \underline{\quad\quad} = \underline{\quad\quad} \end{array}$$

5. Nia used partial sums. Look at her work. Which two numbers did she add?

$$300 + 400 = 700$$

$$20 + 50 = 70$$

$$5 + 3 = 8$$

$$700 + 70 + 8 = 778$$

$$\underline{\quad\quad\quad} + \underline{\quad\quad\quad} = 778$$

6. How can you find the addends by looking at the partial sums?

7. A library lent 186 books on Monday. It lent the same number of books on Tuesday. How many books did it lend in the two days?

Books:

8. Find the sum in a different way. The work in a row is shown. Show it stacked.

$$\begin{array}{l} 352 + 436 = 788 \\ 300 + 400 = 700 \\ 50 + 30 = 80 \\ 2 + 6 = 8 \\ 700 + 80 + 8 = \mathbf{788} \end{array}$$

- 9.** Max adds $457 + 231$. He adds $400 + 200 = 600$ and writes $457 + 231 = 600$. Explain his mistake. What is the correct sum?

Sum: _____

-
- 10.** How can you find $213 + 342 + 125$ using partial sums? Show your work.

Sum: _____

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Together

A	$200 + 300 = 500$; $50 + 10 = 60$; $6 + 8 = 14$ $500 + 60 + 14 = \mathbf{574}$
B	$400 + 100 = 500$; $20 + 60 = 80$; $7 + 5 = 12$ $500 + 80 + 12 = \mathbf{592}$
C	No. 600 is only the hundreds partial sum. Add all three: $600 + 30 + 13 = \mathbf{643}$.

On My Own

1	$300 + 400 = 700$; $50 + 20 = 70$; $8 + 6 = 14$; sum 784 <i>Items 1–4: the two numbers on each line may be in either order.</i>
2	$600 + 200 = 800$; $10 + 70 = 80$; $5 + 3 = 8$; sum 888
3	$500 + 200 = 700$; $30 + 40 = 70$; $9 + 7 = 16$; sum 786
4	$200 + 100 = 300$; $60 + 80 = 140$; $4 + 9 = 13$; sum 453
5	$\mathbf{325 + 453 = 778}$ (either order) <i>First numbers 300, 20, 5 make 325; second numbers 400, 50, 3 make 453.</i>
6	Put the first numbers of each partial-sum equation together to make one addend, and the second numbers to make the other. <i>Accept any explanation that uses place value (hundreds, tens, ones).</i>
7	$186 + 186$: $100 + 100 = 200$; $80 + 80 = 160$; $6 + 6 = 12$. $200 + 160 + 12 = \mathbf{372}$ books
8	Stacked: $352 + 436 \rightarrow 300 + 400 = 700$; $50 + 30 = 80$; $2 + 6 = 8$. Sum 788 <i>The child should write the labels and partial sums in the stacked form, not just copy 788.</i>
9	Max only added the hundreds. He forgot the tens ($50 + 30 = 80$) and ones ($7 + 1 = 8$). $600 + 80 + 8 = \mathbf{688}$.
10	Hundreds: $200 + 300 + 100 = 600$. Tens: $10 + 40 + 20 = 70$. Ones: $3 + 2 + 5 = 10$. $600 + 70 + 10 = \mathbf{680}$.